

Chapter 29 — Software Lifecycle and Change Governance

Field	Value
Status	Revised after whole-book constitutional and cross-chapter audit
Version	0.2
Scope	Treat software, model, schema, configuration, adapter, and infrastructure changes as governed evolutionary transactions rather than informal maintenance.
Constitutional basis	Decisions 33, 37–39, 43, 50–51, 58
Owner review	No unresolved Owner decision after whole-book audit

1. Purpose

Treat software, model, schema, configuration, adapter, and infrastructure changes as governed evolutionary transactions rather than informal maintenance.

2. Authoritative Responsibilities

Component	Authoritative responsibility
Release Manager	Change classification, release channels, provenance, rollout, validation, and retirement
Compatibility Service	Contracts, schemas, coexistence, dependency isolation, and migration gates
Update Verifier	Static, simulation, shadow, canary, and post-update evidence
Rollback Coordinator	Recovery point, reverse migration, prior behavior, and incident linkage

3. Core State and Records

Object	Required content
Change Package	Artifact, type, version, source, signatures, dependencies, permissions, migrations, and risk.
Compatibility Contract	Producer/consumer versions, schema, behavior, deprecation, and test evidence.
Rollout Plan	Environment, cohort, canary, thresholds, observation, promotion, pause, and rollback.
Change Record	Rationale, evidence, authorizations, attempts, outcomes, incidents, and final disposition.

4. Runtime Workflow

1. Classify change as constitutional, kernel, service, adapter, model, schema, configuration, or infrastructure.
2. Verify signed provenance, dependencies, permissions, compatibility, and vulnerability evidence.
3. Create recoverable schema/configuration migrations and a tested rollback or compensation path.
4. Run risk-proportionate static, simulated, shadow, and canary verification.
5. Obtain protected Owner authorization where the change affects constitutional or guarded behavior.
6. Roll out in stages with coexistence where needed and continuously compare expected behavior.
7. Validate after adoption; pause, roll back, compensate, or retire based on evidence.

5. Interfaces and Contracts

Commands, queries, events, records, and evidence are typed, versioned, authenticated, authorized, correlated, and classified. Mutation is accepted only by the authoritative owner. Every external or reality-changing boundary carries identity, policy, authority, provenance, integrity, time, and transaction context.

6. Failure and Recovery

- Unsigned or unverifiable artifact: reject it regardless of claimed urgency.
- Migration partially applies: recover from transaction journal and do not declare the new version active.
- Behavioral drift after model update: quarantine or rollback the provider and preserve comparison evidence.
- Security emergency: accelerate stages only where evidence allows; never disguise governance change as a patch.

7. Constitutional Guarantees

- HAL adopts change because it is proven suitable, not merely newer.
- No maintenance operation may redefine identity, authority, policy, or Constitution by implication.
- Version coexistence and migration preserve continuity and reconstructability.
- Every adopted change has provenance, verification, disposition, and a recoverability record.

8. Prohibited Behaviors

- No component may silently broaden or narrow constitutional authority, reinterpret Owner intent, erase dissent, or promote derived state to authority.
- No failure, degraded dependency, emergency, or optimization may silently change identity, policy, trust, authentication, evidence, or verification rules.
- No model, service, provider, node, Presence, credential, relationship, environment, or network location receives constitutional authority by implication.

9. Observability and Verification

The subsystem records identity, correlation, authority/policy context, inputs, state versions, decisions, limitations, failures, recovery, and outcomes. Verification is risk-scaled, reproducible, and stored as durable evidence. Status and explanation derive from governed state rather than model assertion.

10. Source Alignment and Review

This chapter implements Decisions 33, 37–39, 43, 50–51, 58. Final source-alignment and constitutional disposition is recorded in the batch audit reports; embedded approval text is not treated as review evidence.

Whole-Book Audit Resolution

The following requirements supersede any ambiguous earlier wording in this chapter and are incorporated into this edition:

- Embedded author approval is not conformance evidence. This chapter's constitutional status is determined only by the independent whole-book audit.
- No substantive constitutional or cross-chapter correction was required in this edition.